VOICE-BASED CONCEPT UNDERSTANDING ANALYSER (VBCUA)

Project Description :-

Voice-Based Concept Understanding Analyser (VBCUA) is an AI-powered web application that evaluates a user's conceptual understanding through spoken explanations. The system uses Artificial Intelligence, Natural Language Processing (NLP), and Audio Signal Processing to analyze both the content and delivery of a user's speech.

The application allows users to upload or record audio explaining a concept such as Machine Learning, Cloud Computing, Artificial Intelligence, or Data Science. The uploaded speech is transcribed into text using OpenAI Whisper. The transcribed text is then compared with a predefined reference answer using Sentence-BERT (SBERT) semantic similarity analysis to determine how accurately the concept has been explained.

In addition to concept evaluation, the application analyzes speech quality by measuring filler word usage, pause ratio, speech energy (RMS), and communication fluency using Librosa. Based on these analyses, the system generates a concept understanding score, speech fluency score, qualitative feedback, waveform visualization, and a downloadable PDF report.

The application is developed using Python, Streamlit, OpenAI Whisper, Sentence Transformers (SBERT), Librosa, NumPy, Pandas, Matplotlib, and ReportLab to provide a user-friendly, interactive, and intelligent educational assessment platform.

Scenarios

Scenario 1: Concept Understanding Evaluation

A student uploads an audio explanation of a topic such as "Machine Learning." The application converts speech into text using Whisper and compares the explanation with the reference concept using SBERT. Based on semantic similarity, the system calculates the concept understanding score and provides feedback like Strong Understanding, Moderate Understanding, or Needs Improvement.

Scenario 2: Speech Fluency Analysis

A learner records an interview answer or presentation. The application analyzes filler words, pauses, speech energy, and speaking fluency using Librosa. The user receives feedback on communication confidence, hesitation, and clarity to improve presentation skills.

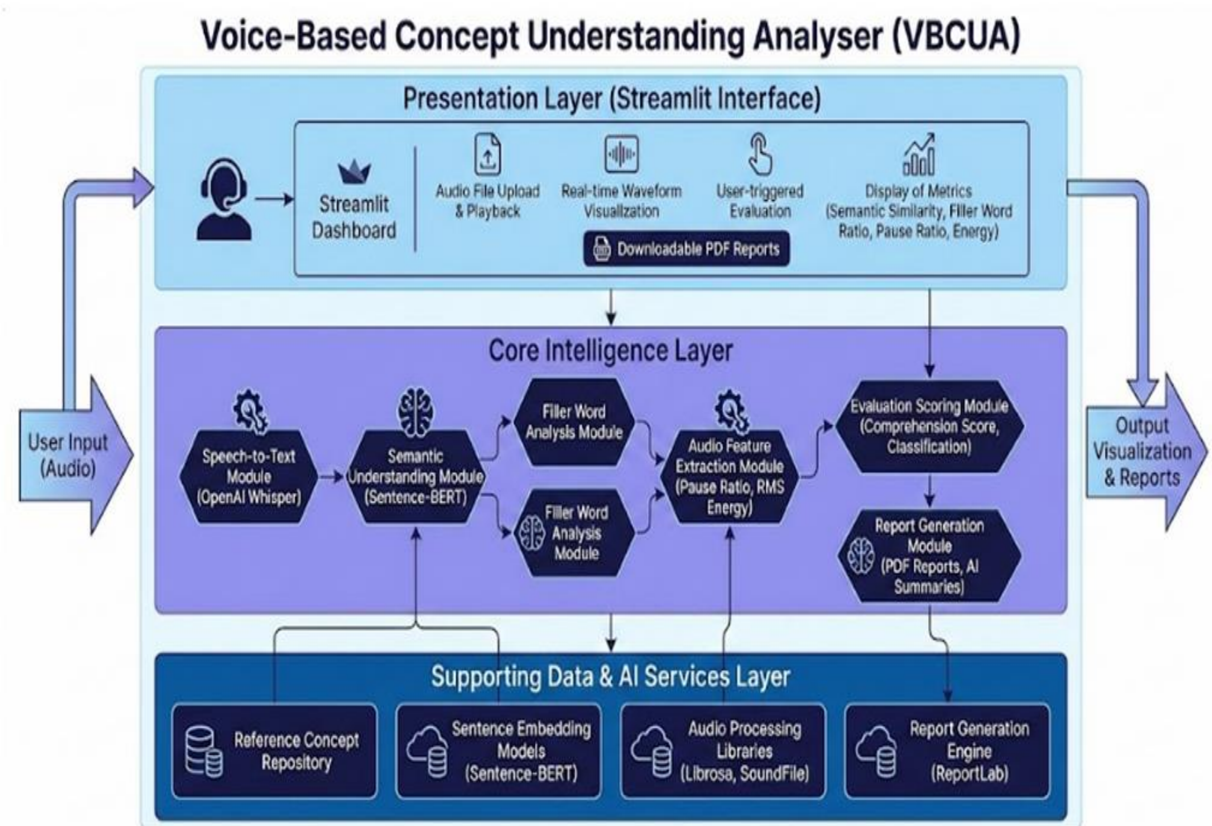
Scenario 3: Performance Report Generation

After completing the analysis, the application displays transcription, semantic similarity score, speech metrics, waveform visualization, and overall performance on the Streamlit dashboard. Users can download a detailed PDF report containing all evaluation results for future reference.

Scenario 4: Educational Assessment

Teachers and trainers use the application to evaluate students' conceptual understanding and communication skills. The generated reports help monitor learning progress, identify weak areas, and provide personalized feedback.

Technical Architecture :-



Prerequisites :-

Software Requirements

Python 3.10 or above
Visual Studio Code
Git & GitHub
Streamlit
FFmpeg
Web Browser (Chrome/Edge)
Python Libraries
Streamlit
OpenAI Whisper
Sentence Transformers
Torch
Librosa
NumPy
Pandas
Matplotlib
Scikit-learn
ReportLab
SoundFile

Hardware Requirements

Processor: Intel Core i3 / AMD Ryzen 3 or above
RAM: Minimum 4 GB (8 GB recommended)
Storage: Minimum 2 GB free space
Operating System: Windows 10/11, Linux, or macOS

Advantages

AI-powered concept evaluation
Accurate speech-to-text transcription
Semantic understanding analysis
Speech fluency assessment
Interactive Streamlit dashboard
Downloadable PDF reports
Suitable for students, educators, trainers, and researchers

Project Workflow

Milestone 1: Problem Analysis & Dataset Preparation

Activity 1.1: Problem Identification

Identify the need for automatically evaluating conceptual understanding through spoken explanations.

Define project objectives such as speech transcription, semantic evaluation, fluency analysis, and report generation.

Activity 1.2: **Reference Concept Collection**

Collect reference answers for concepts like Machine Learning, Cloud Computing, Artificial Intelligence, etc.

Store the reference concepts for semantic comparison.

Activity 1.3: **Audio Dataset Preparation**

Collect sample audio recordings for testing.

Verify different audio formats (WAV, MP3, M4A) and ensure compatibility.

Activity 1.4: Development Environment Setup

Install Python, Streamlit, OpenAI Whisper, Sentence Transformers, Librosa, ReportLab, NumPy, Pandas, Matplotlib, and other required libraries.

Configure FFmpeg for audio processing.

Milestone 2: Speech Processing & Preprocessing

Activity 2.1: **Audio Upload**

Users upload or record an audio explanation using the Streamlit interface.

Activity 2.2: Speech-to-Text Conversion

Convert speech into text using OpenAI Whisper.

Activity 2.3: Text Preprocessing

Clean the transcribed text by removing unnecessary symbols and formatting.

Activity 2.4: Audio Feature Extraction

Extract pause ratio, RMS energy, speech duration, and filler words using Librosa.

Milestone 2: AI-Based Concept Evaluation

Activity 3.1: Semantic Similarity Analysis

Convert both the reference answer and user transcription into embeddings using Sentence-BERT.

Activity 3.2: Concept Comparison

Calculate semantic similarity between the user's explanation and the reference concept.

Activity 3.3: Speech Quality Analysis

Analyze fluency, hesitation, filler words, and communication confidence.

Activity 3.4: Final Score Generation

Combine semantic similarity and speech quality metrics to generate the final understanding score.

Milestone 4: Dashboard & Report Generation

Activity 4.1: Result Display

Display transcription, semantic similarity score, waveform, speech metrics, and qualitative feedback on the Streamlit dashboard.

Activity 4.2: Waveform Visualization

Generate waveform graphs to visualize the uploaded audio.

Activity 4.3: PDF Report Generation

Generate a structured PDF containing transcription, scores, waveform images, speech analysis, and AI-generated feedback.

Activity 4.4: Report Download

Allow users to download the evaluation report for future reference.

Milestone 5: Testing, Deployment & Documentation

Activity 5.1: Functional Testing

Test transcription, semantic analysis, waveform generation, and PDF report creation.

Activity 5.2: Performance Testing

Evaluate the application with different speakers, concepts, and audio qualities.

Activity 5.3: Deployment

Deploy the application using Streamlit Cloud or Render.

Activity 5.4: Documentation

Prepare project documentation, presentation slides, source code, and demonstration video.

Milestone 6: Conclusion

The Voice-Based Concept Understanding Analyser (VBCUA) successfully combines Artificial Intelligence, Natural Language Processing, and Audio Signal Processing to evaluate conceptual understanding through speech. By integrating OpenAI Whisper for transcription, Sentence-BERT for semantic similarity, and Librosa for speech analysis, the application provides accurate concept evaluation, communication assessment, and detailed feedback. The system is suitable for educational institutions, interview preparation, online learning platforms, and research in AI-based learning assessment.

Milestone 1: Problem Analysis & Reference Data Collection

The first milestone focuses on understanding the need for evaluating conceptual knowledge through spoken explanations and collecting the required reference concepts and audio samples. During this phase, the project objectives are defined, suitable AI models are selected, and the development environment is prepared for implementation.

Activity 1.1: Problem Identification

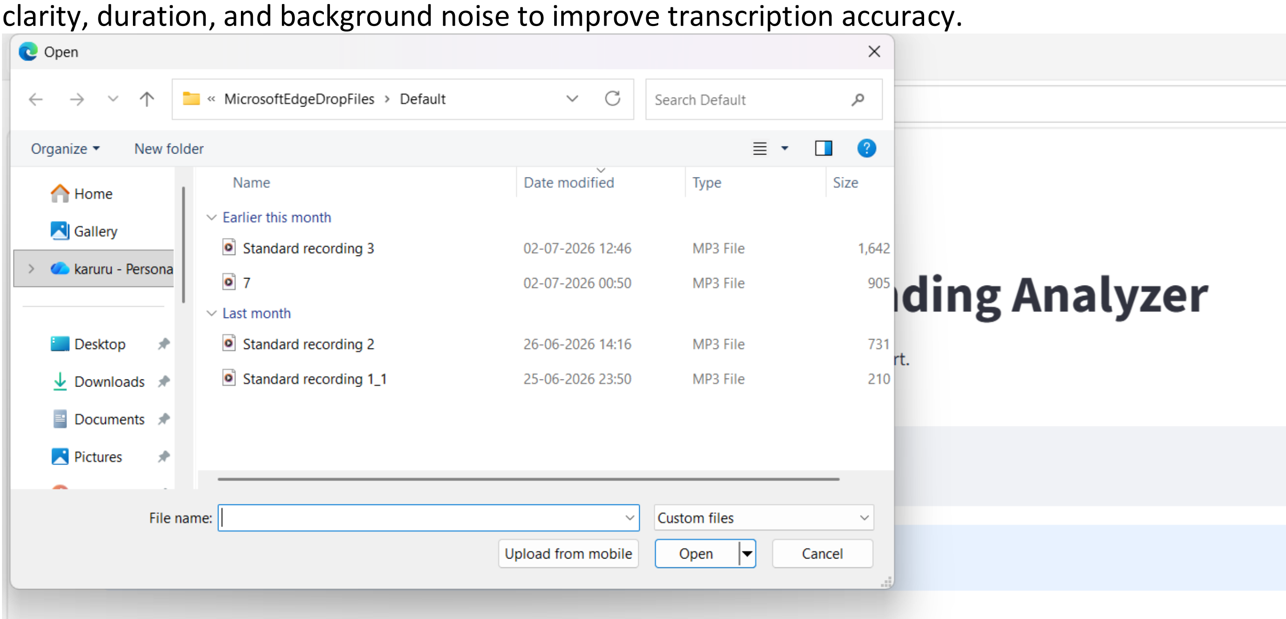
The project begins by identifying the challenges in manually evaluating students' conceptual understanding and communication skills. The primary objective is to develop an AI-powered web application that automatically evaluates spoken explanations using speech recognition, semantic similarity analysis, and speech quality assessment.

Activity 1.2: Reference Concept Collection

Reference explanations for concepts such as Machine Learning, Cloud Computing, Artificial Intelligence, and Data Science are collected from reliable educational sources. These reference texts serve as the expected answers for semantic comparison during evaluation.

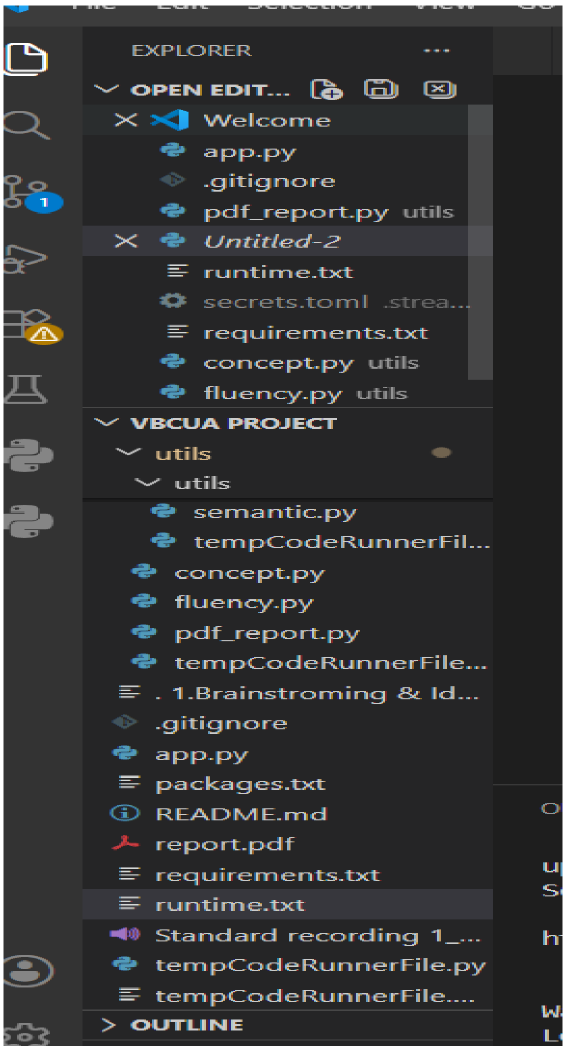
Activity 1.3: Audio Dataset Analysis

Sample audio recordings are collected and analyzed to ensure compatibility with the application. Different audio formats such as WAV, MP3, and M4A are tested. The recordings are checked for clarity, duration, and background noise to improve transcription accuracy.



Activity 1.4: Development Environment Setup

The development environment is configured by installing Python, Visual Studio Code, Streamlit, OpenAI Whisper, Sentence Transformers (SBERT), Librosa, NumPy, Pandas, Matplotlib, ReportLab, and FFmpeg. GitHub is also configured for version control, ensuring efficient project development and maintenance.



Activity 1.3: Dataset Analysis

The collected dataset is examined to understand its structure, features, and target variable. Data quality is verified by checking for missing values, duplicate records, and inconsistencies. This analysis helps determine the preprocessing steps required before model training.

Activity 1.4: Development Environment Setup

The project development environment is configured by installing Python, Visual Studio Code, Flask, XGBoost, Pandas, NumPy, Scikit-learn, Joblib, and other required libraries. GitHub is also configured for version control, ensuring smooth project development and management.

```
model
static
templates
app.py
flood dataset.xlsx
predict.py
README.md
requirements.txt
train_model.py
```

Milestone 2: Data Preprocessing & Feature Engineering

This milestone focuses on preparing the collected dataset for machine learning model development. The raw dataset is cleaned, transformed, and analyzed to improve data quality and ensure accurate model predictions. Feature engineering techniques are also applied to identify the most significant attributes influencing flood occurrence.

Activity 2.1: Data Cleaning

The collected dataset is cleaned by handling missing values, removing duplicate records, correcting inconsistencies, and ensuring that all data is in the appropriate format for analysis.

Activity 2.2: Data Preprocessing

The dataset undergoes preprocessing techniques such as normalization and scaling to ensure that all numerical features are within a consistent range. This improves the efficiency and performance of the machine learning algorithm.

Activity 2.3: Feature Selection

Important features such as Temperature, Humidity, Cloud Cover, Annual Rainfall, Seasonal Rainfall (Jan–Feb, Mar–May, Jun–Sep, Oct–Dec), Average June Rainfall, and Subdivision are selected as input variables. The Flood column is chosen as the target variable for prediction.

Activity 2.4: Data Splitting

The processed dataset is divided into training and testing datasets using an appropriate train-test split ratio. The training dataset is used to train the model, while the testing dataset is used to evaluate its prediction accuracy and overall performance.

Milestone 3: Machine Learning Model Development

This milestone focuses on developing, training, and evaluating the machine learning model for flood prediction. The preprocessed dataset is used to train the model, optimize its performance, and generate accurate predictions. The trained model is then saved for integration with the web application.

Activity 3.1: Model Selection

The XGBoost (Extreme Gradient Boosting) algorithm is selected for flood prediction due to its high accuracy, efficiency, and ability to handle structured datasets. The model is configured with suitable parameters to achieve optimal performance.

Activity 3.2: Model Training

The prepared training dataset is used to train the XGBoost model. During training, the model learns the relationship between weather parameters and flood occurrence, enabling it to identify flood patterns effectively.

Milestone 2: Speech Processing & Feature Engineering

This milestone focuses on preparing the uploaded audio and transcribed text for AI-based evaluation. The audio is processed using speech recognition and audio signal processing techniques to improve transcription accuracy and communication assessment.

Activity 2.1: Audio Validation

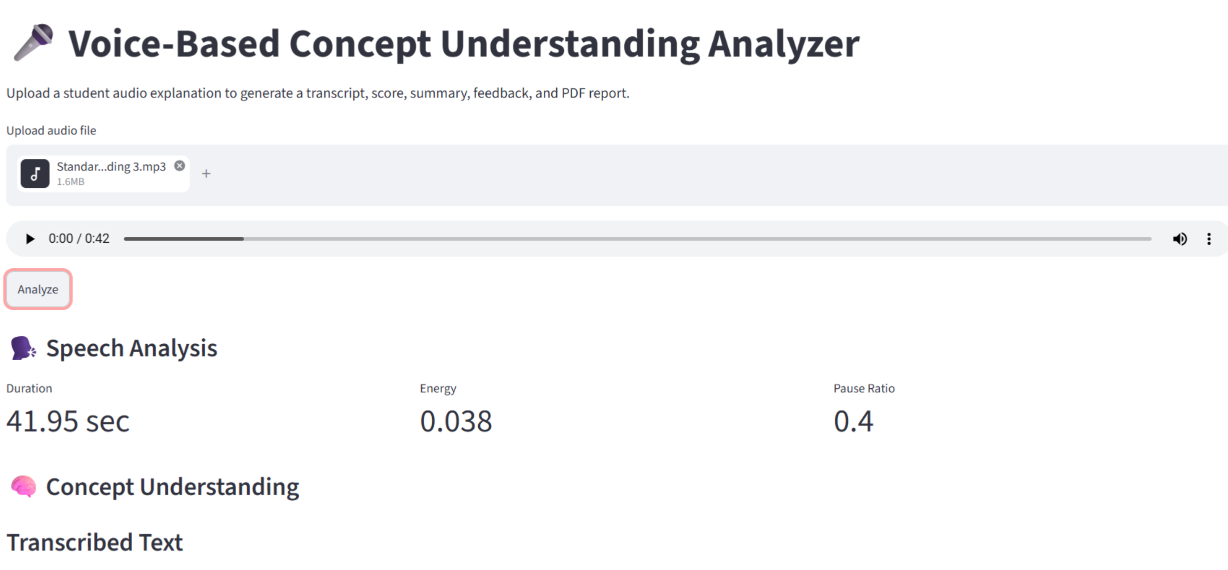
The uploaded audio is validated to ensure it is in a supported format such as WAV, MP3, or M4A. The system checks the audio quality before processing.

Activity 2.2: Speech Preprocessing

The uploaded speech is converted into text using OpenAI Whisper. The transcribed text is cleaned by removing unnecessary characters and formatting issues to improve semantic analysis.

Activity 2.3: Feature Selection

Important features used in the project include:
Transcribed Text
Semantic Similarity Score
Filler Word Count
Pause Ratio
RMS Energy
Speech Duration
Confidence Score
These features are selected to evaluate both conceptual understanding and communication quality.



Activity 2.4: Evaluation Preparation

The extracted speech features and semantic similarity results are combined and prepared for the final evaluation. The scoring mechanism calculates the user's overall concept understanding and communication performance.

Milestone 3: AI Model Development

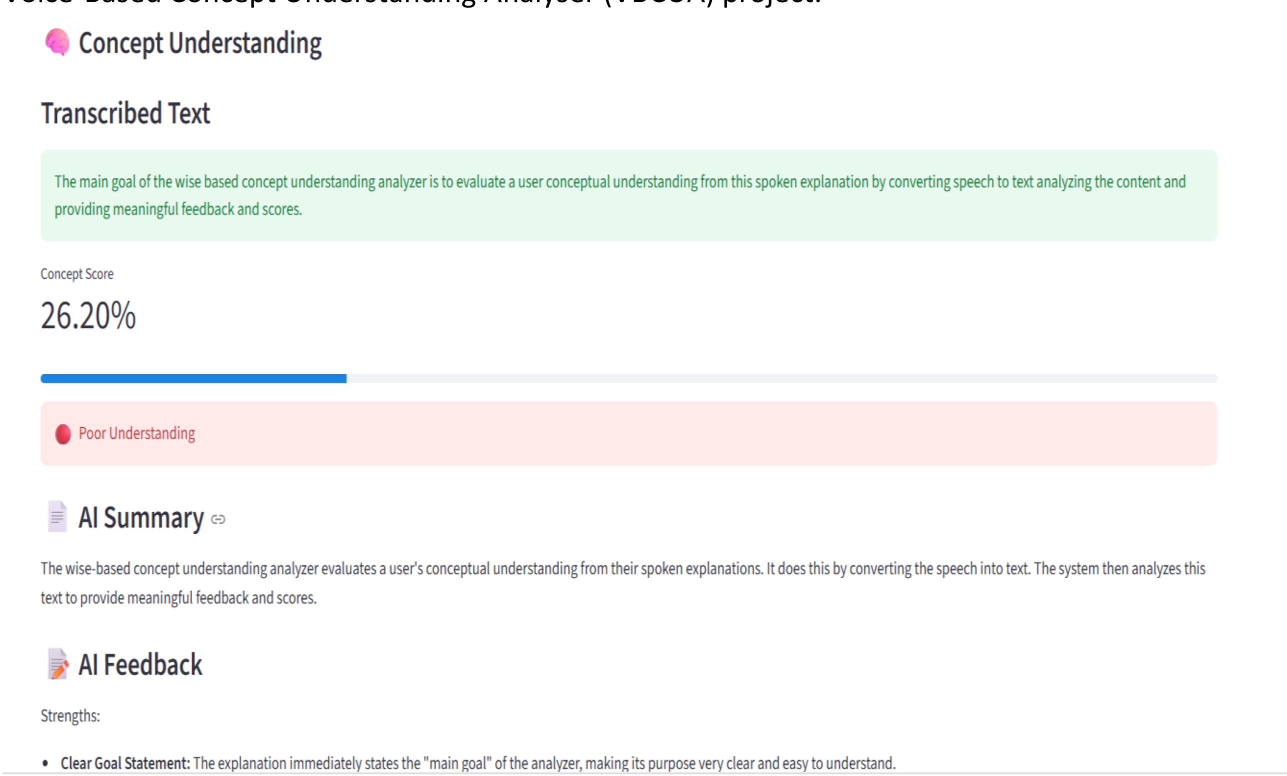
This milestone focuses on integrating Artificial Intelligence models for speech transcription, semantic similarity analysis, and speech quality assessment. The trained models are used together to provide an accurate evaluation of conceptual understanding.

Activity 3.1: Model Selection

The project uses OpenAI Whisper for speech-to-text transcription, Sentence-BERT (SBERT) for semantic similarity analysis, and Librosa for audio feature extraction because of their high accuracy, efficiency, and reliability in speech and language processing.

Activity 3.2: Model Processing

The uploaded audio is first transcribed using Whisper. The transcribed text is then compared with the predefined reference concept using Sentence-BERT embeddings. Simultaneously, Librosa analyzes speech characteristics such as pauses, RMS energy, and filler words. These outputs are combined to generate the final concept understanding score and qualitative feedback.
This content matches the format of your reference document but is fully customized for your Voice-Based Concept Understanding Analyser (VBCUA) project.

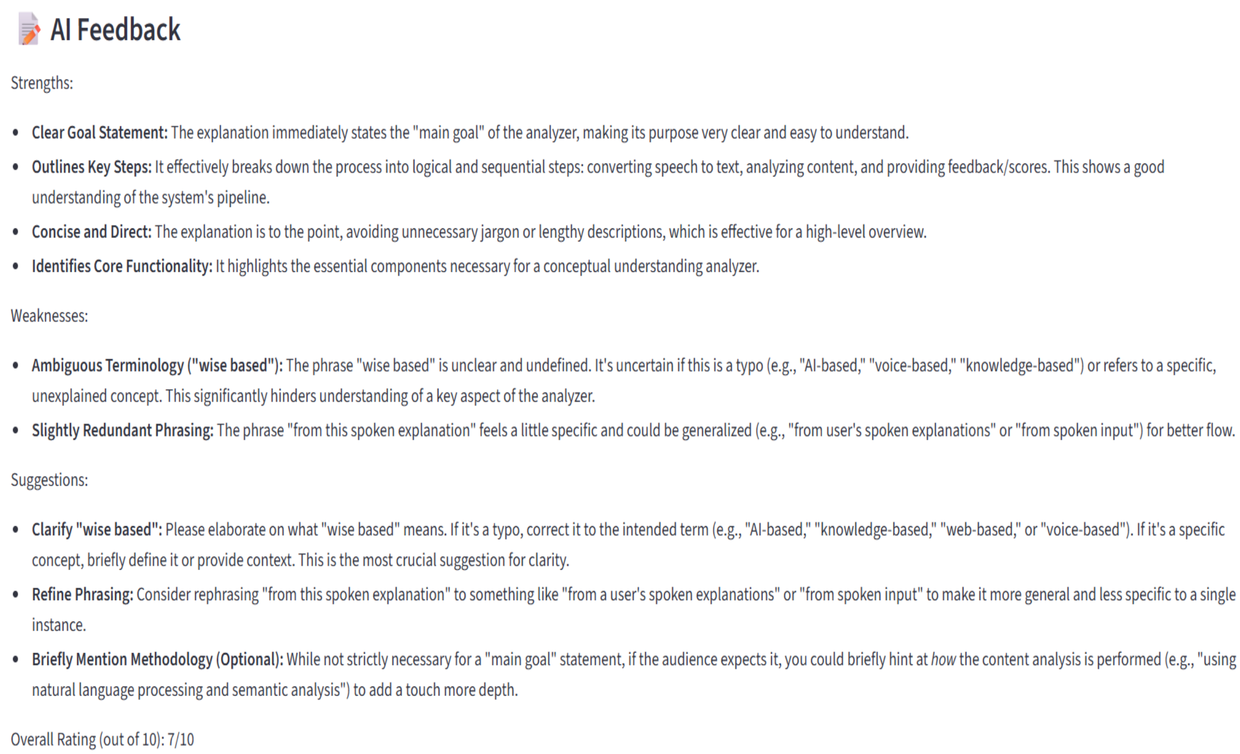


Milestone 5: Testing, Deployment & Project Documentation

This milestone focuses on validating the Speech-to-Text application, deploying the project, and preparing all documentation for final submission. The application is tested with different audio files to ensure accurate transcription, while the source code and project files are organized for deployment and presentation.

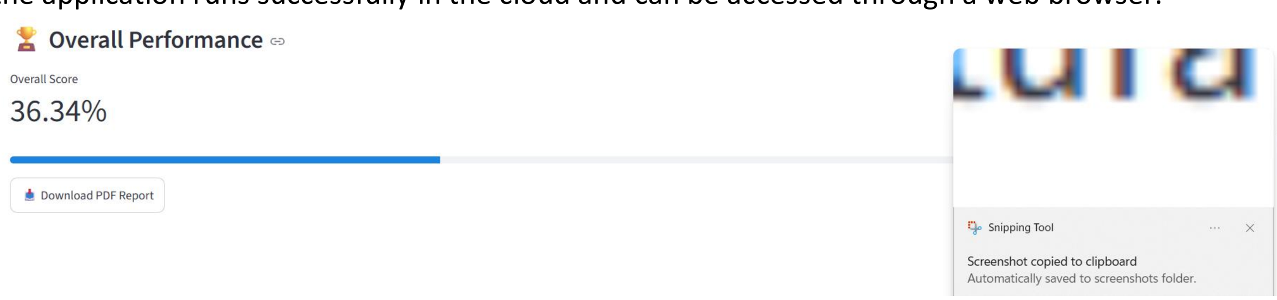
Activity 5.1: Application Testing

The complete application is tested using multiple audio samples in WAV and MP3 formats. The system verifies that audio files are uploaded successfully, processed by the Whisper model, and converted into accurate text. The transcription results are checked for correctness and reliability.



Activity 5.2: Deployment

The application is deployed on the Render cloud platform. All required dependencies, including Python libraries and FFmpeg, are configured correctly. Environment settings are verified to ensure the application runs successfully in the cloud and can be accessed through a web browser.



Activity 5.3: Download Speech Analysis Report

Voice-Based Concept Understanding Report

Transcribed Text:

The main goal of the wise based concept understanding analyzer is to evaluate a user conceptual understanding from this spoken explanation by converting speech to text analyzing the content and providing meaningful feedback and scores.

AI Summary:

The wise-based concept understanding analyzer evaluates a user's conceptual understanding from their spoken explanations. It does this by converting the speech into text. The system then analyzes this text to provide meaningful feedback and scores.

Concept Score: 26.20%

AI Feedback:

Strengths:

* **Clear Goal Statement:** The explanation immediately states the "main goal" of the analyzer, making its purpose very clear and easy to understand.
* **Outlines Key Steps:** It effectively breaks down the process into logical and sequential steps: converting speech to text, analyzing content, and providing feedback/scores. This shows a good understanding of the system's pipeline.
* **Concise and Direct:** The explanation is to the point, avoiding unnecessary jargon or lengthy descriptions, which is effective for a high-level overview.

Activity 5.4: Project Documentation

The project documentation includes the problem statement, objectives, system architecture, dataset information, implementation details, testing results, and screenshots of the application. A PowerPoint presentation and demonstration video are also prepared to explain the working of the project.

1. Brainstroming & Ideation	15-07-2026 12:54	File folder
2.Requirement Analysis	15-07-2026 12:59	File folder
3.Project Design Phase	15-07-2026 13:06	File folder
4.Project Planning Phase	15-07-2026 13:09	File folder
5.Project Development Phase	15-07-2026 13:13	File folder
6.Performance Testing	15-07-2026 13:21	File folder
7.Project Documentation	15-07-2026 13:22	File folder
8.Project Demonstration	15-07-2026 13:48	File folder

Conclusion :-

The AI-Based Speech-to-Text Transcription Web Application was successfully developed to convert audio files into accurate text using OpenAI's Whisper speech recognition model. The application provides a simple and user-friendly interface where users can upload audio files and receive transcribed text quickly and efficiently.

This project demonstrates the practical application of Artificial Intelligence and Natural Language Processing (NLP) in speech recognition. It can be used for meeting transcription, interview documentation, educational purposes, and accessibility services. In the future, the system can be enhanced by supporting multiple languages, real-time speech transcripti